

A Three-Sheet Inverse-Monodromy Criterion for the Quadratic Smirnov Uniqueness Condition

Candidate partial-result packet for arXiv:2601.04146

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Status: substantial partial result, likely valid; pending specialist review
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1 Source problem

Fricain, Grivaux, Ostermann, and Yakubovich study commutants and embeddability of Toeplitz operators with smooth symbols in [1]. If Ω is a component of $\sigma(T_F) \setminus F(\mathbb{T})$ with $|\text{wind}_F(\Omega)| = 2$, put

$$\Gamma = \partial\Omega \cap \partial\Omega_1, \quad \zeta = 1/F^{-1}$$

on the regular arcs of the parametrized curve. Condition (1) of their Theorem 6.7 is

$$u + \zeta v + \zeta^2 w = 0 \quad \text{a.e. on } \Gamma, \quad u, v, w \in E^1(\Omega) \implies u = v = w = 0. \quad (1)$$

Immediately after the theorem, the authors state that they do not know a geometric condition implying (1). Their earlier Property (P) treats the corresponding two-term condition [2], but does not supply the three-term criterion.

2 The inverse-cover condition

The relevant geometric datum is not merely the planar set $F(\mathbb{T})$ but the sheet structure of the parametrized reciprocal inverse. This distinction is unavoidable: Example 6.6 of the source gives the same planar curve square-root and cube-root parametrizations with opposite commutant behavior.

Definition 1. *Let Ω be a bounded Smirnov domain with piecewise analytic boundary and let $\Gamma \subset \partial\Omega$ contain an analytic arc of positive length. We say that (Ω, ζ) has a bounded finite inverse continuation of sheet degree d if there are*

- (i) *a connected Riemann surface X and a finite proper holomorphic map $\pi : X \rightarrow \Omega$ of degree d ;*
- (ii) *a function $\Psi \in H^\infty(X)$ which takes d distinct values on a generic fiber of π ;*
- (iii) *a positive-length analytic subarc $\gamma \subset \Gamma$ and a one-sided local section s of π over an interior collar of γ such that the non-tangential boundary values of $\Psi \circ s$ equal ζ a.e. on γ .*

The continuation has full boundary compatibility if, on almost every regular subarc of Γ , ζ is the boundary value of some local sheet of Ψ .

Thus d counts distinct analytic values, not merely points in an artificially enlarged covering. For $\zeta(\lambda) = \lambda^{1/d}$, take $X = \mathbb{D}$, $\pi(x) = x^d$, and $\Psi(x) = x$.

These two examples show that it might be difficult to find conditions of a geometric nature on the curve $F(\mathbb{T})$ implying that the commutant of T_F consists of multiplication operators. We thus finish this section by presenting an *analytic* condition, inspired by Example 6.6, implying that the assumptions of Proposition 6.4 are satisfied. Under this condition, T_F is embeddable into a C_0 -semigroup on H^p if and only if 0 belongs to the unbounded component of $\mathbb{C} \setminus \text{int}(\sigma(T_F))$.

Theorem 6.7. *Let $p > 1$, and let F be a symbol satisfying the assumptions (H1), (H2) and (H3). Suppose that for every connected component Ω of $\sigma(T_F) \setminus F(\mathbb{T})$, the following condition holds:*

- (1) *if $|\text{wind}_F(\Omega)| = 2$, then for any $u, v, w \in E^1(\Omega)$ we have*

$$u + \zeta v + \zeta^2 w = 0 \text{ a.e. on } \partial\Omega \cap \partial\Omega_1 \implies u = v = w = 0 \text{ on } \Omega.$$
- (2) *if $k := |\text{wind}_F(\Omega)| > 2$, for any $u, v \in E^1(\Omega)$ we have*

$$u + \zeta v = 0 \text{ a.e. on } \partial\Omega \cap \partial\Omega_{k-1} \implies u = v = 0 \text{ on } \Omega.$$

Then $\{M_\lambda\}' = \{M_a; a \in \text{Mult}(E_F^q)\}$, i.e. the commutant of M_λ consists of multiplication operators only.

As a consequence, T_F is embeddable into a C_0 -semigroup on H^p if and only if 0 belongs to the unbounded component of $\mathbb{C} \setminus \text{int}(\sigma(T_F))$.

Condition (2) of Theorem 6.7 has appeared already in [19], where a geometric condition (called Property (P)) implying that (2) holds was given. As already mentioned above, we do not know any geometric condition implying that the analytic condition (1) of Theorem 6.7 holds.

Proof. The fact that all multiplication operators belong to the commutant of M_λ is clear. Consider now $A \in \{M_\lambda\}'$, and let us prove that there exists $a \in \text{Mult}(E_F^q)$

Figure 1: Theorem 6.7 and the stated missing geometric condition in [1, p. 30].

3 Proof intuition

The boundary relation is first converted, by Privalov uniqueness on a small one-sided collar, into a holomorphic identity on one inverse sheet. Lifting to X and using connectedness propagates that identity to every sheet. Over a regular value λ , the same polynomial in one variable must therefore vanish at all d distinct values of Ψ in the fiber. A polynomial of degree at most $r < d$ cannot do so unless all its coefficients vanish.

The converse is just as rigid. If $d \leq r$, multiply the linear factors $t - \Psi(x)$ over a fiber. The elementary symmetric coefficients descend to bounded holomorphic functions on Ω , including across branch values, and the resulting nonzero degree- d polynomial annihilates every boundary sheet. This makes the sheet threshold sharp.

4 Finite-sheet classification

Theorem 2. *Let $r \geq 0$, and suppose (Ω, ζ) has a bounded finite inverse continuation of sheet degree d in the sense of Definition 1. If $d > r$, then*

$$\sum_{k=0}^r a_k \zeta^k = 0 \quad \text{a.e. on } \Gamma, \quad a_0, \dots, a_r \in E^1(\Omega), \quad (2)$$

implies $a_0 = \dots = a_r = 0$ on Ω .

If full boundary compatibility holds and $d \leq r$, the uniqueness conclusion fails. Consequently, within the fully compatible bounded finite-continuation class, degree- r boundary uniqueness holds if and only if $d > r$.

Proof. Put

$$P(\lambda, t) = \sum_{k=0}^r a_k(\lambda) t^k.$$

Shorten the seed arc γ from Definition 1 if necessary, and choose a piecewise analytic one-sided collar $W \subset \Omega$ whose artificial boundary is compactly contained in Ω . Let $\psi = \Psi \circ s$ on W . It is bounded.

Every $a_k \in E^1(\Omega)$ belongs to the Nevanlinna class $\mathcal{N}(\Omega)$. Restricting a quotient of bounded holomorphic functions shows that $a_k|_W \in \mathcal{N}(W)$. Since ψ is bounded and the Nevanlinna class is a field, $P(\lambda, \psi(\lambda)) \in \mathcal{N}(W)$. Relation (2) makes its non-tangential boundary values zero on a positive-length arc, so the standard Privalov boundary uniqueness theorem for Nevanlinna-class functions gives

$$P(\lambda, \psi(\lambda)) = 0 \quad (\lambda \in W). \quad (3)$$

The function

$$x \mapsto P(\pi(x), \Psi(x))$$

is holomorphic on X . Equation (3) says that it vanishes on an open subset. Since X is connected, it vanishes identically. For every regular value λ at which Ψ separates the fiber, the polynomial $P(\lambda, t)$ therefore has the d distinct roots

$$\{\Psi(x) : x \in \pi^{-1}(\lambda)\}.$$

If $d > r$, this polynomial is identically zero. Thus every a_k vanishes on the nonempty open set of such regular values, and the identity theorem gives $a_k = 0$ throughout Ω .

For the converse, initially at regular values define

$$Q(\lambda, t) = \prod_{x \in \pi^{-1}(\lambda)} (t - \Psi(x)) = t^d + c_{d-1}(\lambda)t^{d-1} + \cdots + c_0(\lambda).$$

The coefficients are symmetric in the fiber, hence are single-valued and holomorphic away from the branch values. If $\|\Psi\|_\infty \leq M$, then $|c_{d-j}| \leq \binom{d}{j} M^j$. The removable-singularity theorem extends each coefficient through the discrete branch-value set, so $c_j \in H^\infty(\Omega) \subset E^1(\Omega)$. Full boundary compatibility implies

$$Q(\lambda, \zeta(\lambda)) = 0 \quad \text{a.e. on } \Gamma.$$

This is a nonzero relation of degree $d \leq r$; append zero higher coefficients when $d < r$. Uniqueness therefore fails. $\square$

Corollary 3 (Sharp three-sheet criterion). *If (Ω, ζ) has a bounded inverse continuation with at least three sheets, then the source condition (1) holds. Under full boundary compatibility in the finite-sheet class, it holds exactly when $d \geq 3$.*

Proof. Apply Theorem 2 with $r = 2$. For $d = 2$, the explicit obstruction is

$$\zeta^2 - (\psi_1 + \psi_2)\zeta + \psi_1\psi_2 = 0,$$

where the symmetric sum and product descend to $H^\infty(\Omega)$. The one-sheet case already has a nonzero linear relation. $\square$

Combining Corollary 3 with Theorem 6.7 of [1] shows that, when no component of winding modulus greater than two occurs, the three-sheet condition forces the commutant of the model operator to consist only of multipliers and yields the paper's embeddability criterion.

5 An explicit infinite family

For each integer $m \geq 3$, let $a_m = 2\pi(m-1)/m$ and define

$$F_m(e^{i\theta}) = \begin{cases} 2 - m + (m-1)e^{-im\theta/(m-1)}, & 0 \leq \theta \leq a_m, \\ e^{-im\theta}, & a_m \leq \theta \leq 2\pi. \end{cases} \quad (4)$$

At $\theta = 0, a_m, 2\pi$, the pieces have the same value 1 and matching first derivative $-im$. Both pieces have constant nonzero speed m ; the derivative is globally Lipschitz. Thus F_m is $C^{1,1}$ as a function on $\mathbb{T}$ and satisfies the source's smoothness hypothesis (H1) for every required Hölder exponent below one.

The image consists of two clockwise circles: the circle centered at $2-m$ of radius $m-1$, and the unit circle. They are internally tangent only at 1. Consequently (H2) holds, all winding numbers are nonpositive, and

$$\text{wind}_{F_m}(\lambda) = \begin{cases} -2, & |\lambda| < 1, \\ -1, & |\lambda - (2-m)| < m-1 \text{ and } |\lambda| > 1, \\ 0, & \text{outside the outer circle.} \end{cases}$$

On the boundary of the doubly wound component $\Omega = \mathbb{D}$,

$$\zeta(\lambda) = \exp\left(\frac{1}{m} \text{Log}_{(0,2\pi)} \lambda\right).$$

It has the continuation model

$$X = \mathbb{D}, \quad \pi(x) = x^m, \quad \Psi(x) = x,$$

of sheet degree m . Corollary 3 proves (1) for every $m \geq 3$. The member $m = 3$ is the cube-root example already proved directly in Example 6.6 of the source; the classification supplies the structural explanation and all $m \geq 4$ at once.

6 Verification and limitations

The current verdict is **likely valid** (confidence 91/100).

Check	Finding
Boundary-to-interior step	Since $E^1(\Omega) \subset \mathcal{N}(\Omega)$, restriction to the collar and multiplication by a bounded inverse sheet stay in $\mathcal{N}(W)$; Privalov boundary uniqueness applies.
Continuation	The relation is lifted to one connected surface; no comparison between unrelated local sectors is made.
Branch values	Symmetric coefficients are bounded near the discrete branch set and extend removably.
Sharpness	The fiber polynomial gives an explicit nonzero relation whenever $d \leq r$.
Family sanity check	<code>code/verify_family.py</code> checks junction values, first derivatives, representative winding numbers, and the three-root Vandermonde determinant for $m = 3, \dots, 20$. This is not a proof.

The theorem requires bounded finite analytic continuation of the reciprocal inverse. It does not classify arbitrary symbols whose inverse continuation has infinitely many sheets, natural boundaries, poles, or no continuation beyond a collar. It is also a condition on the sheet geometry of the *parametrized* curve, not on the unparametrized planar image alone. The source’s square-root/cube-root pair shows why the latter distinction is essential.

For these reasons the packet is conservatively classified as a substantial partial result. A specialist should decide whether a finite inverse-cover condition counts as the geometric condition requested by the source. Within the bounded finite-continuation class, Theorem 2 is a full sharp classification.

7 Bounded novelty search

On 9 August 2026, the run’s lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:2601.04146, the displayed three-term relation, inverse monodromy, and finite-sheet variants. The only relevant prior attempt, on arXiv:2502.03303, concerns different hypercyclicity questions. Official-arXiv web searches used the source title together with “monodromy,” “reciprocal inverse monodromy commutant,” “Smirnov quadratic relation,” and the displayed relation. They found the source paper but no paper stating this criterion or classification. This bounded search supports but does not guarantee novelty.

References

- [1] E. Fricain, S. Grivaux, M. Ostermann, and D. Yakubovich, *Embedding of Toeplitz operators with smooth symbols into strongly continuous semigroups*, arXiv:2601.04146 (2026), especially Example 6.6 and Theorem 6.7.
- [2] E. Fricain, S. Grivaux, and M. Ostermann, *Hypercyclicity of Toeplitz operators with smooth symbols*, arXiv:2502.03303 (2025), especially Property (P) and Proposition 4.9.